VARUNA 2047 — ForestWatch Intelligence Report

Region: WAR | **Type:** District | **Generated:** 2026-10-27 13:54 UTC | **Classification:** Research Grade

1. EXECUTIVE SUMMARY & CANOPY INTEGRITY

This evidence dossier presents monitored canopy changes and satellite observations for WAR. Under the Forest Survey of India (ISFR) benchmark, baseline forest extent is recorded at **1,157.6 hectares**. During the observation window, measured tree-cover loss area totaled **236.21 hectares** (20.41% of baseline).

2. TREE-LOSS METRICS & DEFENSIBLE ESTIMATION

Metric Indicator	Measured Value	Scientific Standard
Measured Tree-Cover Loss Area	236.21 ha	Hansen GFC v1.11 / Sentinel-2 dNDVI (10m)
Percentage Canopy Change	20.41%	100 * (Loss Area / Baseline Forest Area)
Estimated Tree Stems Lost	N/A	FSI NFI Average Stem Density per Hectare (+/- 25%)
Change Trajectory	STABLE	Forest loss rate remains within historical baseline bounds (+/- 15%
Baseline Sentinel-2 Scene	S2B_44QLE_20190129_0_L2A...	ESA Copernicus BOA Surface Reflectance
Recent Sentinel-2 Scene	S2B_44QLE_20240113_0_L2A...	ESA Copernicus BOA Surface Reflectance

3. SCIENTIFIC TRANSPARENCY & LIMITATION DISCLOSURES

Important Methodological Note: Tree-cover loss detected: 0.0 hectares. Individual trees lost cannot be reliably determined from the available satellite resolution.
Spaceborne multispectral imagery (10m–30m) captures canopy-cover disturbance rather than individual tree stems. Tree counts cannot be defensibly calculated without sub-meter crown-detection sensors or field inventory plots. Canopy-loss pixels do NOT imply clearcut logging or illegal felling without ground enforcement records.